

STANDARD F

H. pylori Ag FIA

STANDARD™ F *H. pylori* Ag FIA
PLEASE READ INSTRUCTIONS CAREFULLY
BEFORE YOU PERFORM THE TEST

SD BIOSENSOR

KIT CONTENTS



Test device
(individually in a foil pouch
with desiccant)



Sterile swab



Extraction buffer tube



Nozzle cap



Instructions for use

SPECIMEN COLLECTION AND PREPARATION

■ Feces

1. Fecal specimen should be collected in a clean and dry container and collected at any time of the day may be used.
2. Urine should be excluded from the fecal specimen.
3. Fresh fecal specimen should be used in this test. Do not use any transport media to store and transport the fecal specimen.
4. Fecal specimen may be stored at room temperature (15-25 °C / 59-77 °F) for up to 2 days and in a refrigerator at (2-8 °C / 36-46 °F) for up to 4 days.
5. For over 4 days storage, specimens may need to be frozen at -40 ± 2 °C / -40 ± 3.6 °F for up to 2 weeks.
6. It should be brought to room temperature prior to use.

TEST PROCEDURE

■ Preparation

1. Carefully read instructions for using STANDARD F *H. pylori* Ag FIA.
2. Take out the test device in foil pouch.
* Check the valid expiry date at the back of the foil pouch. Do not use if the expiry date has passed.



Foil pouch



Test device



Yellow : Valid
Green : Invalid



The control line appears on the membrane of the test device after use.



Before Use After Use
Do not write on the barcode or damage the barcode of the test device.

■ Specimen Preparation

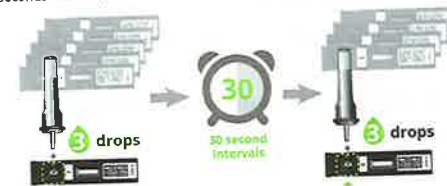
1. Poke the 4 different areas of the feces specimen using the sterile swab to collect a portion of feces (about 40-70 mg). In case of liquid specimen, soak the sterile swab completely in the liquid fecal specimen.



1. Take the test device out of the foil pouch and place it on a flat and dry surface. Write a specimen information on the label of test device.
2. Prepare extracted specimens.
3. Prepare test devices depending on the workload.



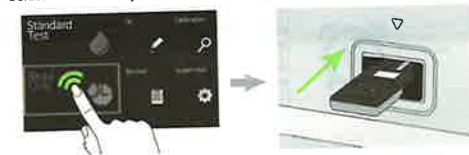
4. Apply 3 drops of extracted specimen into test devices in sequence at about 30 seconds intervals.



5. Incubate the test device for 10 minutes outside of the analyzer.



6. Select the 'Read Only' mode on the screen, and insert test devices in sequence.



7. The analyzer will automatically scan and display the test result immediately.

INTERPRETATION OF TEST RESULT

■ Interpretation

Result	COI (Cutoff index) value	Interpretation
Positive	COI ≥ 1.0	<i>H. pylori</i> Ag Positive
Negative	COI < 1.0	<i>H. pylori</i> Ag Negative
Invalid	COI value is not displayed	Retest should be performed with a new test device and a new patient specimen

The test result of a specimen is given either as Positive (+) / Pos (+) or Negative (-) / Neg (-) with a COI (cut off index) value. The COI is a numerical representation of the measured fluorescence signal.

QUALITY CONTROL

■ Calibration

The calibration set test of STANDARD F analyzer should be conducted according to the analyzer's manual.

[When to use calibration set]

1. Before using the analyzer for the first time.
2. When you drop the analyzer.
3. Whenever you do not agree with the final result.
4. When you want to check the performance of an analyzer and test device.

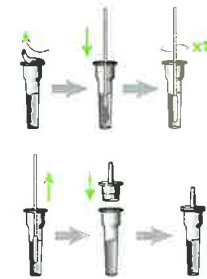
[How to use calibration set]

Calibration set test is a required function that ensures optimal performance by checking the internal analyzer optics and functions.

1. Select the 'Calibration' in main menu.

Swab amount of feces to show in the picture below. Excessive fecal amount may induce a false positive result and slow migration.

- Take off the top of an Extraction buffer tube and insert the swab into the tube. Swirl the swab at least ten times into the extraction buffer tube to dissolve the specimen.
- Remove the swab and dispose of the used swab in accordance with your biohazard waste disposal protocol. Tightly press down the nozzle cap onto the tube.



Do not use the extraction buffer tube without the nozzle cap. It affects the result value.

Analysis of specimen

'STANDARD TEST' mode

STANDARD F200 and F2400 Analyzer

- Take the test device out of the foil pouch and place it on a flat and dry surface. Write patient information on the label of test device.

STANDARD F2400 Analyzer	'Workplace' → 'Run Test' → Insert patient ID and / or operator ID on the analyzer
STANDARD F200 Analyzer	'STANDARD TEST' mode → Insert patient ID and/or operator ID

- Insert the test device to the test slot of the analyzer. When inserting the test device to the analyzer, the analyzer read the barcode data, and check the test device is valid.
- Apply 3 drops of mixed specimen to the specimen well in the test device holding the prepared extraction buffer tube upside down.
- After applying the specimen, immediately press the 'TEST START' button.
- The analyzer will automatically display the test result after 10 minutes.



STANDARD F analyzer automatically calibrate and identify the optical performance through measuring the membrane of the test device whenever the test is conducted in 'Standard Test' mode. If 'EEE' message displays on the screen, it means that the analyzer has a problem, so check with test device. Contact the SD BIOSENSOR local distributor if the 'EEE' message still appears.

EXPLANATION AND SUMMARY

Introduction

Helicobacter pylori is a small, spiral-shaped bacterium that lives on the surface of the stomach and duodenum. The organism is very common, infected at least half of the world's population. *H. pylori* infection is associated with the etiology of a variety of gastrointestinal diseases, including non-ulcer dyspepsia, duodenal and gastric ulcer and active and chronic gastritis. Studies also suggest an association of *H. pylori* infection with stomach cancer; the role of *H. pylori* and the factors involved in the development of these diseases are still under investigation. Both invasive and non-invasive methods are used to diagnose *H. pylori* infection in patients with symptoms of gastrointestinal disease. Specimen dependent and costly invasive diagnostic methods include gastric or duodenal biopsy followed by urease testing, culture or histologic staining. Noninvasive techniques include the urea breath test, serological methods and *Helicobacter pylori* stool antigen test. STANDARD F *H. pylori* Ag FIA provides the significantly fast, easy and accurate system to detect the *H. pylori* antigens in the human fecal specimen. It is essential for the reliable clinical diagnosis of *H. pylori* infection and enables supportive treatment decisions.

Intended use

STANDARD F *H. pylori* Ag FIA is a qualitative immunoassay to detect *H. pylori* Ag in the human fecal specimen. The test is for *in vitro* diagnostic use and is intended as an aid to early diagnosis of *H. pylori* Ag infection. This is intended for professional use, only for an initial screening test.

Test Principle

STANDARD F *H. pylori* Ag FIA is based on the immunofluorescence technology with STANDARD F Analyzer manufactured by SD BIOSENSOR to measure *H. pylori* antigen in human fecal specimens. The specimen from human fecal should be processed for the preparation using the components of the STANDARD F *H. pylori* Ag FIA. After applying the specimen mixture to the test device, the complex will be formed on the membrane as the result of the antigen-antibody reaction. The intensity of the fluorescence light produced on the membrane is scanned by the STANDARD F Analyzer. STANDARD F Analyzers can analyze the *H. pylori* antigen of the clinical specimen based on a pre-programmed algorithms and display the test result on the screen.

KIT STORAGE AND STABILITY

Store the kit at 2-30 °C / 36-86 °F, out of direct sunlight. Kit materials are stable until the expiration date printed on the outer box. Do not freeze.

WARNINGS AND PRECAUTIONS

- For *in vitro* diagnostic use only.
- Carefully follow the instructions and procedures described in this instruction while testing.
- STANDARD F *H. pylori* Ag FIA should be performed with STANDARD F Analyzers.
- STANDARD F *H. pylori* Ag FIA test device should remain in its original sealed pouch until ready to use. Do not use if the pouch is damaged or the seal is broken.
- STANDARD F *H. pylori* Ag FIA is single use only. Do not re-use it.
- Do not use an artificial materials as specimen.
- Place the analyzer on a flat surface when in use.
- Wash your hands in warm and soapy water. Rinse well and dry completely before testing.

PERFORMANCE CHARACTERISTICS

CLINICAL PERFORMANCE

A total of 98 laboratory specimen has been prospectively analyzed during three non-consecutive days and studied in parallel with STANDARD™ F *H. pylori* Ag FIA and Reference assay (CLIA method). Equivocal results obtained with reference assay has been considered as discordant and then retested. The test results are shown below;

		Reference Assay (CLIA method)		
		Positive	Negative	Equivocal
STANDARD™ F <i>H. pylori</i> Ag FIA	Positive	47	3	2
	Negative	2	42	2

Relative Sensitivity: 94.00% (47/49) (83.45% - 98.75%)
Relative Specificity: 95.45% (42/45) (84.53% - 99.44%)
(Positive Predictive Value: 95.92% / Negative Predictive Value: 93.33%)

ANALYTICAL PERFORMANCE

- Analytical Sensitivity (Limit of Detection) The analytical sensitivity (Limit of Detection)

- Cross Reactivity STANDARD™ F *H. pylori* Ag FIA do

#	Potential
1	
2	
3	



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ANALYTICAL PERFORMANCE

1. Analytical Sensitivity (Limit of Detection)

The analytical sensitivity [Limit of Detection (LoD)] of STANDARD™ F H. pylori Ag FIA is shown below;

Limit of Detection (CFU/mL)		
5.0 x 10 ⁶		

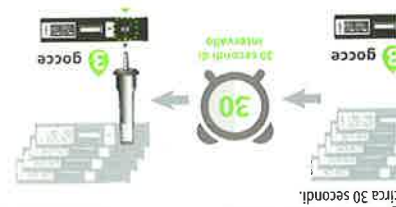
2. Cross Reactivity		
STANDARD™ F H. pylori Ag FIA does not be affected by those potential cross-reactive microorganisms and viruses		
#	Potential Cross-reacting substance	Concentration
1	Adenovirus type 1	2.0x10 ⁶
2	Adenovirus type 2	2.0x10 ⁶
3	Adenovirus type 3	2.0x10 ⁶



sullo schermo la modalità "Read Only" (Solo lettura) e inserire il



r 10 minuti il dispositivo diagnostico fuori dall'analizzatore.



gocce di campione estratto nel dispositivo diagnostico in sequenza a

1. I risultati dei test devono essere considerati insieme ad altri dati clinici e precauzioni e l'interpretazione dei risultati del test stesso.
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3. Per una maggiore precisione dello stato immunologico, sono raccomandati test di follow-up aggiuntivi usando altri metodi di laboratorio.
4. La mancata osservanza della procedura del test e dell'interpretazione dei risultati può influire negativamente sulle prestazioni del test e/o produrre risultati non validi.
5. Se la concentrazione ematica di streptavidina è molto alta possono verificarsi interferenze.
6. Il test non è influenzato da livelli (bilirubina ≤257 µmol/l) e biotina (<300 ng/ml).
7. Nel caso di pazienti che ricevono dosi elevate di biotina (>5 mg/giorno), i campioni devono essere prelevati almeno 8 ore dopo l'ultima somministrazione di biotina.

7. התאמת המערכת למטרות הלימודים